

Indian Statistical Institute
First Semester Examinations: 2024-25

Course Name: M. Math, 1st year
Subject Name : Topology I
Maximum Marks: 40, Duration: Two hours
Date: 9.9.2024, 2:30 PM - 4:30 PM

- Answer as many questions as you can.
- Maximum marks is 40.
- You may use any results proved in class. Any other results (including those in homework problem sets) require proof.

1. a) Let $f : X \rightarrow Y$ be a function where X is a set and Y a topological space. Consider the collection $\mathfrak{T} = \{f^{-1}(U) \mid U \text{ is open in } Y\}$. Prove that $\mathfrak{T}$ is a topology on X . 5
- b) Consider the function $f : \mathbb{R} \rightarrow S^1$ which sends t to $e^{2\pi it}$. Check that the induced topology on $\mathbb{R}$ using the construction in a) is not the same as the usual topology. 5
2. Let $Y \subset X$ and assume that both X and Y are connected. Suppose that $X \setminus Y = A \cup B$ is a separation. Prove that $Y \cup A$ and $Y \cup B$ are connected. 8
3. Let A and B be disjoint compact subsets of a Hausdorff space X . Show that there exists disjoint open sets U and V containing A and B respectively. 8
4. a) Let X be a normal space. Let $A \subset X$ and $B \subset X$ be disjoint closed subsets. Prove that there exists open subsets U, V with $A \subset U$, $B \subset V$ such that $\bar{U} \cap \bar{V} = \emptyset$. 4
- b) Prove that a closed subset of a normal space is normal. 4
5. a) Let $X = [0, 1) \cup (3, 4] \subset \mathbb{R}$.
i) Show that X is locally compact. 2
ii) Prove that the one-point compactification of X is homeomorphic to the closed interval $[0, 1]$. 4
b) Let $Y = (0, 1) \cup (3, 4)$. Prove that the one-point compactification of Y is not homeomorphic to $[0, 1]$. 4